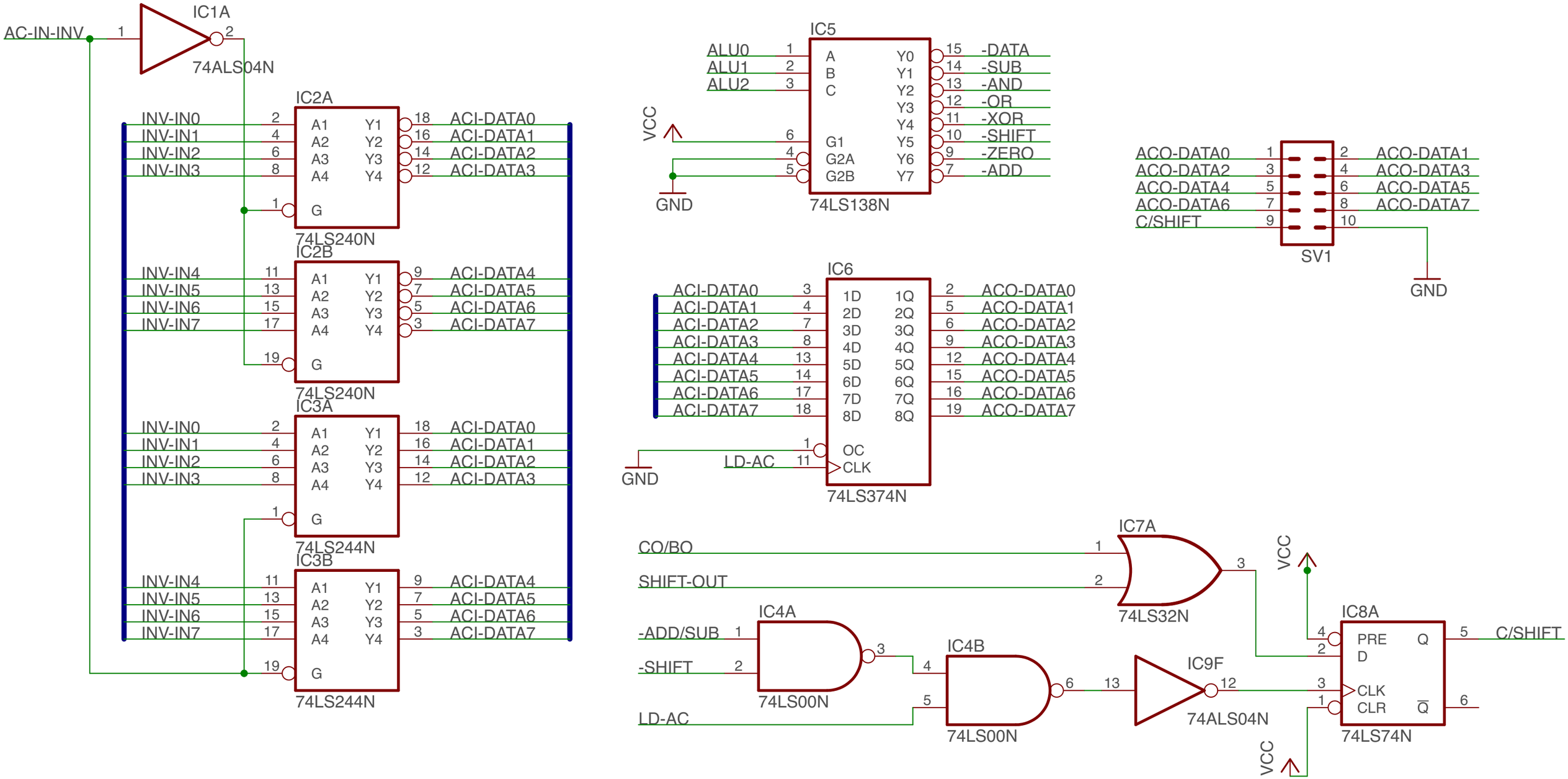
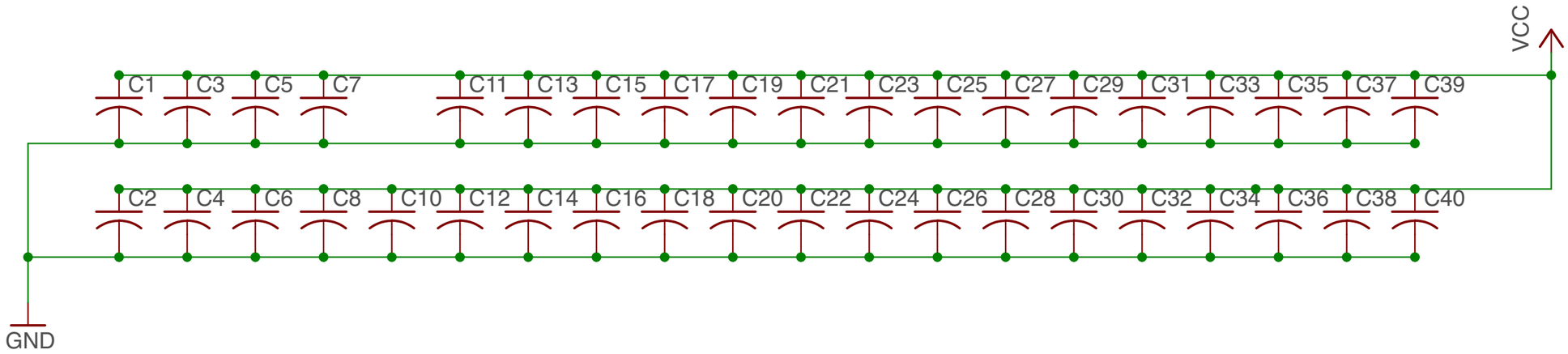
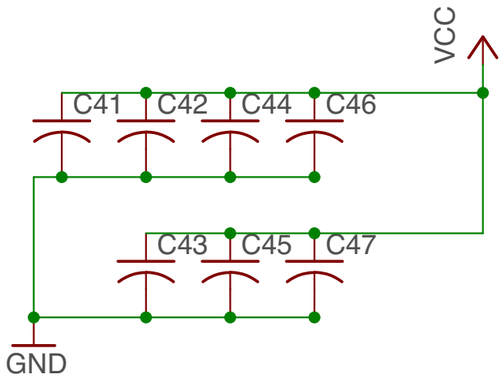
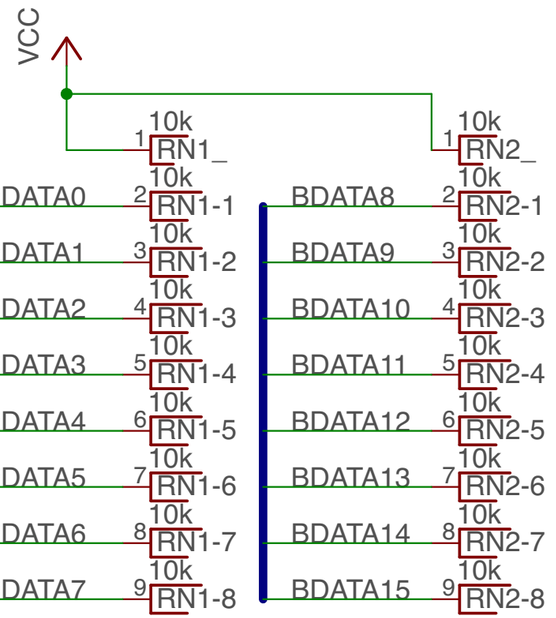

AC-IN-INV
IC1A
74ALS04N
IC2A
INV-IN0
INV-IN1
INV-IN2
INV-IN3
ACI-DATA0
ACI-DATA1
ACI-DATA2
ACI-DATA3
74LS240N
IC2B
INV-IN4
INV-IN5
INV-IN6
INV-IN7
ACI-DATA4
ACI-DATA5
ACI-DATA6
ACI-DATA7
74LS240N
IC3A
74LS244N
IC3B
74LS244N
IC5
ALU0
ALU1
ALU2
VCC
GND
-DATA
-SUB
-AND
-OR
-XOR
-SHIFT
-ZERO
-ADD
74LS138N
IC6
ACI-DATA0
ACI-DATA1
ACI-DATA2
ACI-DATA3
ACI-DATA4
ACI-DATA5
ACI-DATA6
ACI-DATA7
ACO-DATA0
ACO-DATA1
ACO-DATA2
ACO-DATA3
ACO-DATA4
ACO-DATA5
ACO-DATA6
ACO-DATA7
LD-AC
GND
74LS374N
ACO-DATA0
ACO-DATA2
ACO-DATA4
ACO-DATA6
C/SHIFT
ACO-DATA1
ACO-DATA3
ACO-DATA5
ACO-DATA7
SV1
GND
IC7A
CO/BO
SHIFT-OUT
74LS32N
IC4A
-ADD/SUB
-SHIFT
74LS00N
IC4B
LD-AC
74LS00N
IC9F
74ALS04N
VCC
VCC
IC8A
PRE
D
CLK
CLR
Q
Q
C/SHIFT
74LS74N


VCC
C1
C3
C5
C7
C11
C13
C15
C17
C19
C21
C23
C25
C27
C29
C31
C33
C35
C37
C39
C2
C4
C6
C8
C10
C12
C14
C16
C18
C20
C22
C24
C26
C28
C30
C32
C34
C36
C38
C40
GND


VCC
C41
C42
C44
C46
C43
C45
C47
GND


VCC
10k
1
RN1_
10k
1
RN2_
BDATA0
2
RN1-1
BDATA8
2
RN2-1
BDATA1
3
RN1-2
BDATA9
3
RN2-2
BDATA2
4
RN1-3
BDATA10
4
RN2-3
BDATA3
5
RN1-4
BDATA11
5
RN2-4
BDATA4
6
RN1-5
BDATA12
6
RN2-5
BDATA5
7
RN1-6
BDATA13
7
RN2-6
BDATA6
8
RN1-7
BDATA14
8
RN2-7
BDATA7
9
RN1-8
BDATA15
9
RN2-8

IC9A
I/D-REG-OUT 1
2 -I/D-REG-OUT
74ALS04N

IC9D
I/D-REG-UP 9
8 -I/D-REG-UP
74ALS04N

IC9C
I/D-REG-DN 5
6 -I/D-REG-DN
74ALS04N

RESET

IC9B
I/D-REG-LD 3
4 -I/D-REG-LD
74ALS04N

IC10
BDATA0 15
BDATA1 1
BDATA2 10
BDATA3 9
A B C D QA QB QC QD
5 UP
4 DN
11 LD
14 CLR
CO 12
BO 13
74LS192N

IC11
BDATA4 15
BDATA5 1
BDATA6 10
BDATA7 9
A B C D QA QB QC QD
5 UP
4 DN
11 LD
14 CLR
CO 12
BO 13
74LS192N

IC12
BDATA8 15
BDATA9 1
BDATA10 10
BDATA11 9
A B C D QA QB QC QD
5 UP
4 DN
11 LD
14 CLR
CO 12
BO 13
74LS192N

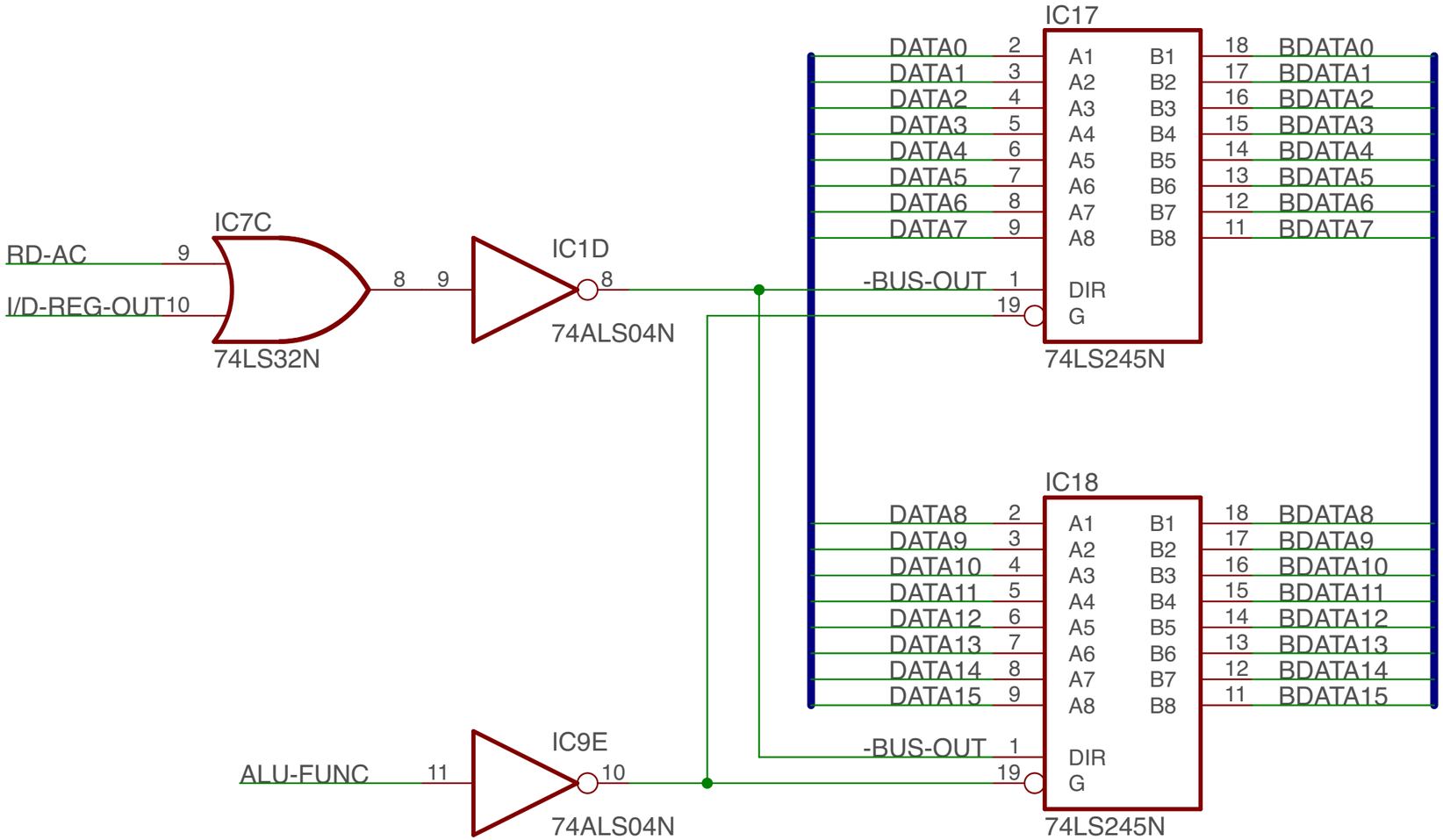
IC13
BDATA12 15
BDATA13 1
BDATA14 10
BDATA15 9
A B C D QA QB QC QD
5 UP
4 DN
11 LD
14 CLR
CO 12
BO 13
74LS192N

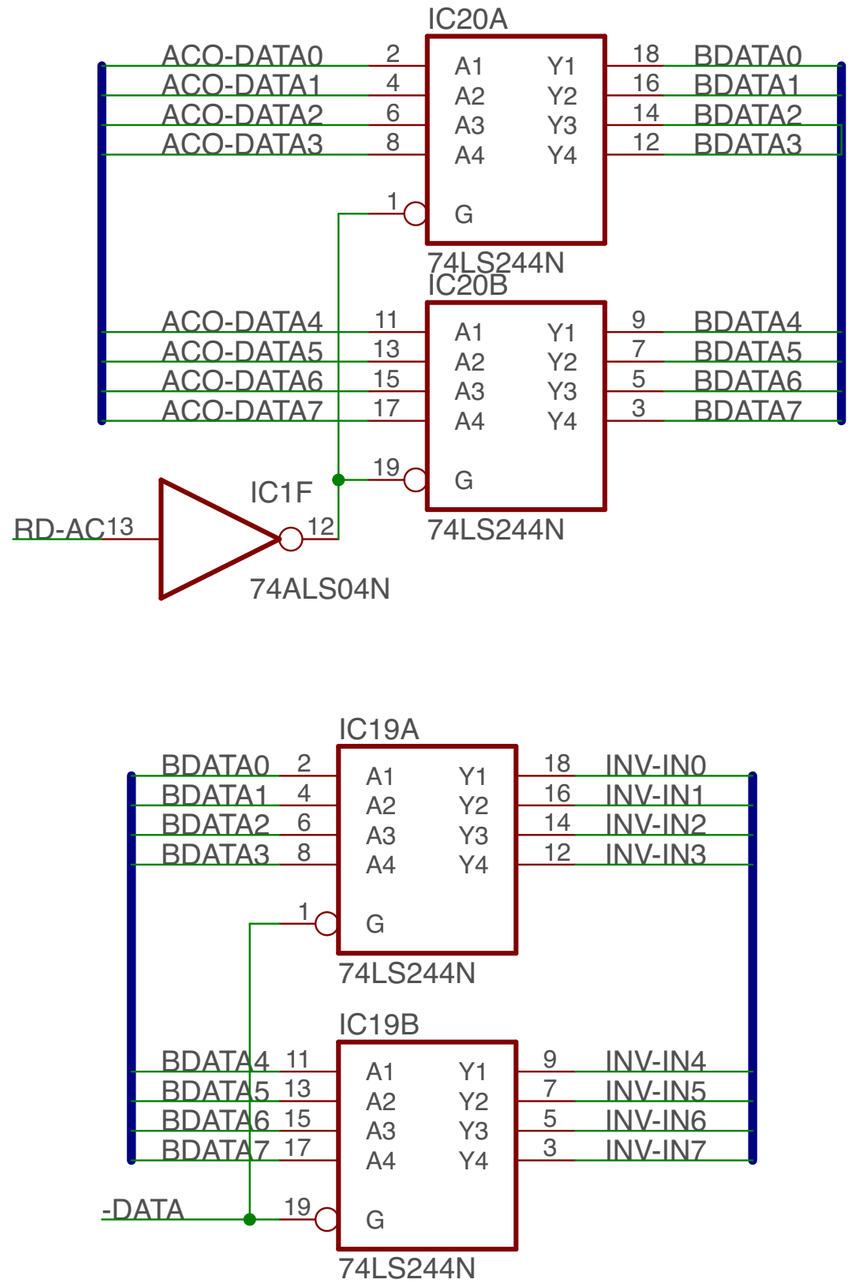
IC14A
2 4 6 8 A1 A2 A3 A4 Y1 Y2 Y3 Y4 18 16 14 12
BDATA0 BDATA1 BDATA2 BDATA3
1 G
74LS244N

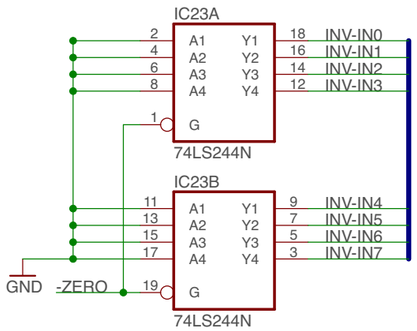
IC14B
11 13 15 17 A1 A2 A3 A4 Y1 Y2 Y3 Y4 9 7 5 3
BDATA4 BDATA5 BDATA6 BDATA7
19 G
74LS244N

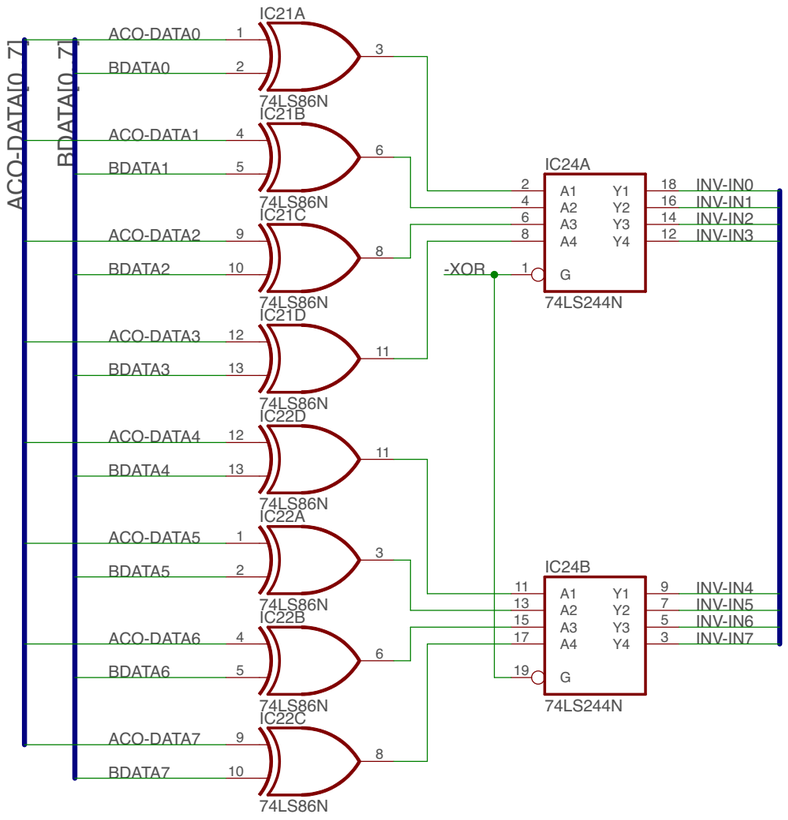
IC15A
2 4 6 8 A1 A2 A3 A4 Y1 Y2 Y3 Y4 18 16 14 12
BDATA8 BDATA9 BDATA10 BDATA11
1 G
74LS244N

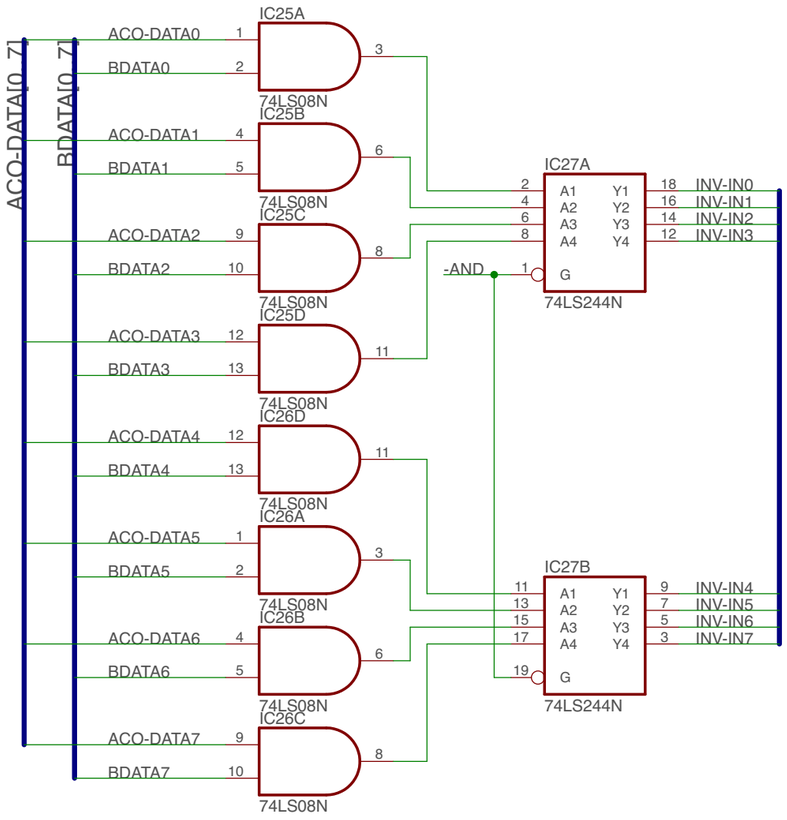
IC15B
11 13 15 17 A1 A2 A3 A4 Y1 Y2 Y3 Y4 9 7 5 3
BDATA12 BDATA13 BDATA14 BDATA15
19 G
74LS244N

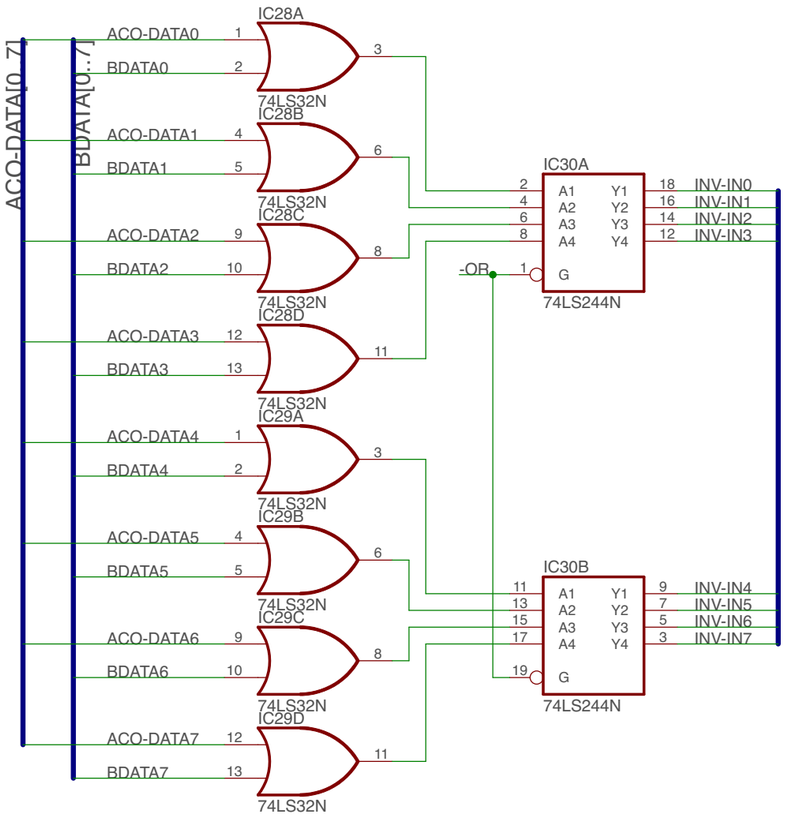

IC7C
RD-AC
9
I/D-REG-OUT
10
74LS32N
8
9
IC1D
8
74ALS04N
IC17
DATA0
DATA1
DATA2
DATA3
DATA4
DATA5
DATA6
DATA7
2
3
4
5
6
7
8
9
A1
A2
A3
A4
A5
A6
A7
A8
B1
B2
B3
B4
B5
B6
B7
B8
18
17
16
15
14
13
12
11
BDATA0
BDATA1
BDATA2
BDATA3
BDATA4
BDATA5
BDATA6
BDATA7
-BUS-OUT
1
DIR
19
G
74LS245N
IC18
DATA8
DATA9
DATA10
DATA11
DATA12
DATA13
DATA14
DATA15
BDATA8
BDATA9
BDATA10
BDATA11
BDATA12
BDATA13
BDATA14
BDATA15
-BUS-OUT
74LS245N
IC9E
ALU-FUNC
11
10
74ALS04N

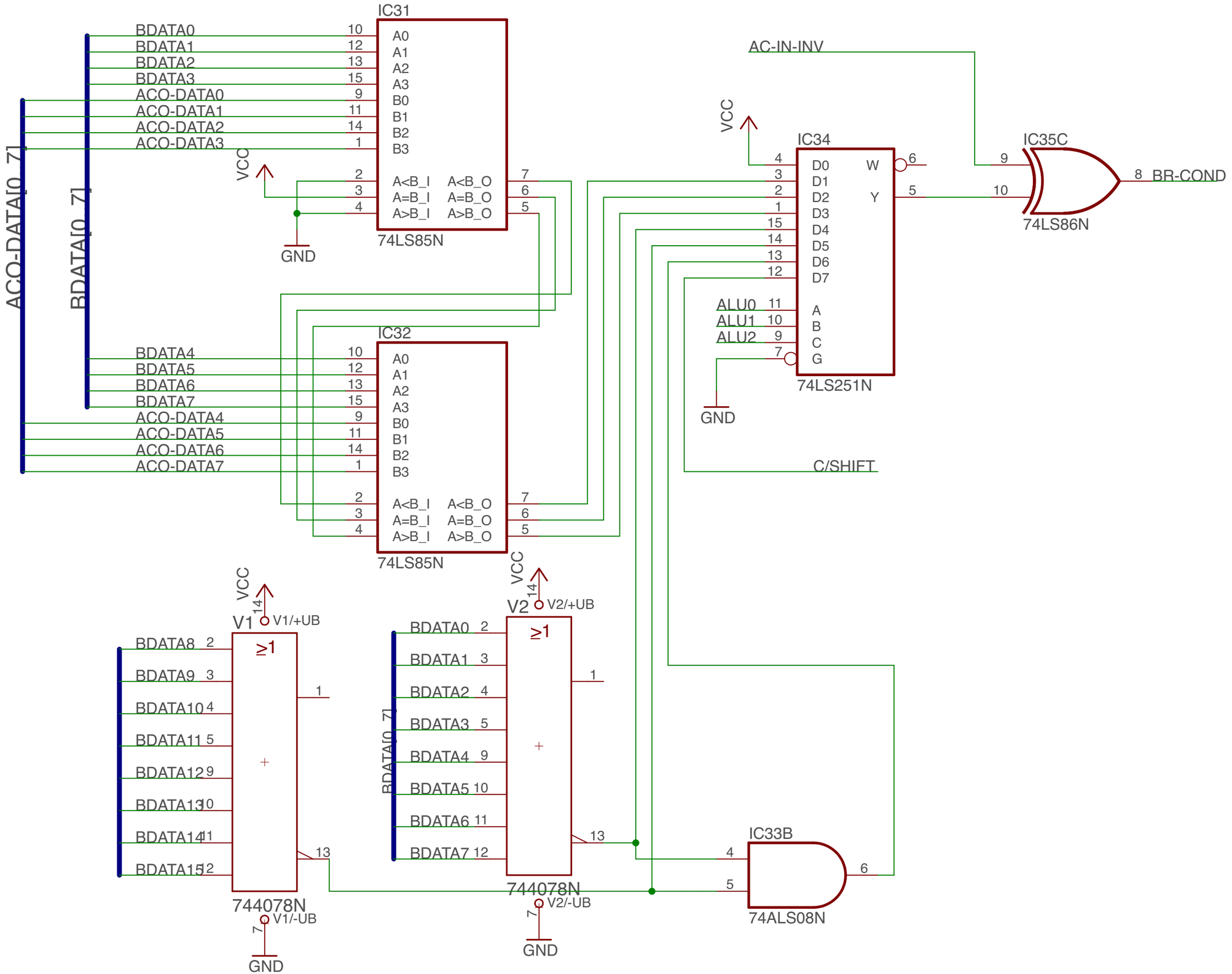

IC20A
ACO-DATA0
ACO-DATA1
ACO-DATA2
ACO-DATA3
2
4
6
8
A1
A2
A3
A4
Y1
Y2
Y3
Y4
18
16
14
12
BDATA0
BDATA1
BDATA2
BDATA3
1
G
74LS244N
IC20B
ACO-DATA4
ACO-DATA5
ACO-DATA6
ACO-DATA7
11
13
15
17
9
7
5
3
BDATA4
BDATA5
BDATA6
BDATA7
19
IC1F
RD-AC
13
12
74ALS04N
IC19A
BDATA0
BDATA1
BDATA2
BDATA3
INV-IN0
INV-IN1
INV-IN2
INV-IN3
IC19B
BDATA4
BDATA5
BDATA6
BDATA7
INV-IN4
INV-IN5
INV-IN6
INV-IN7
-DATA

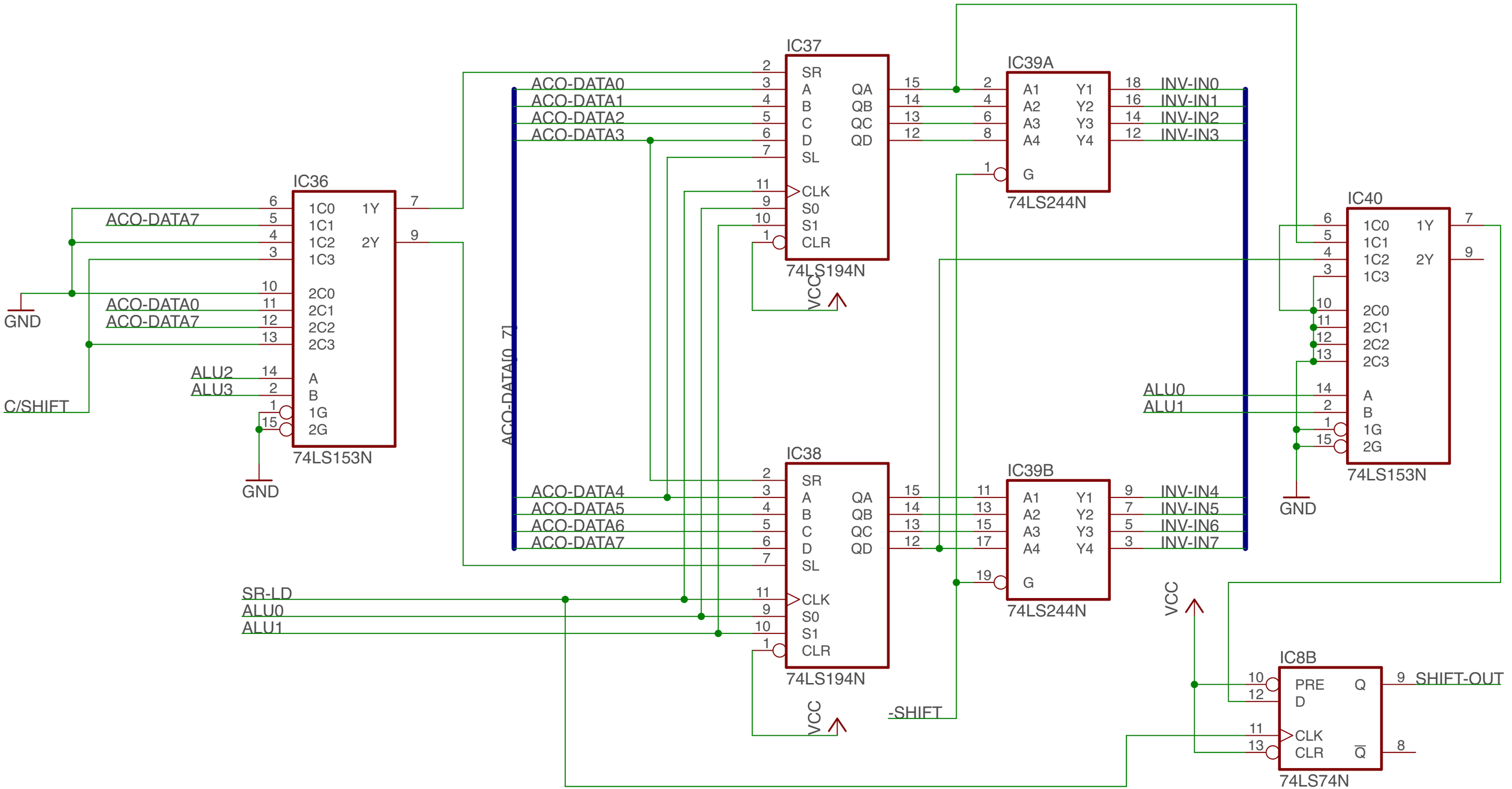

IC23A
A1 Y1
A2 Y2
A3 Y3
A4 Y4
G
74LS244N
IC23B
GND
-ZERO
INV-IN0
INV-IN1
INV-IN2
INV-IN3
INV-IN4
INV-IN5
INV-IN6
INV-IN7

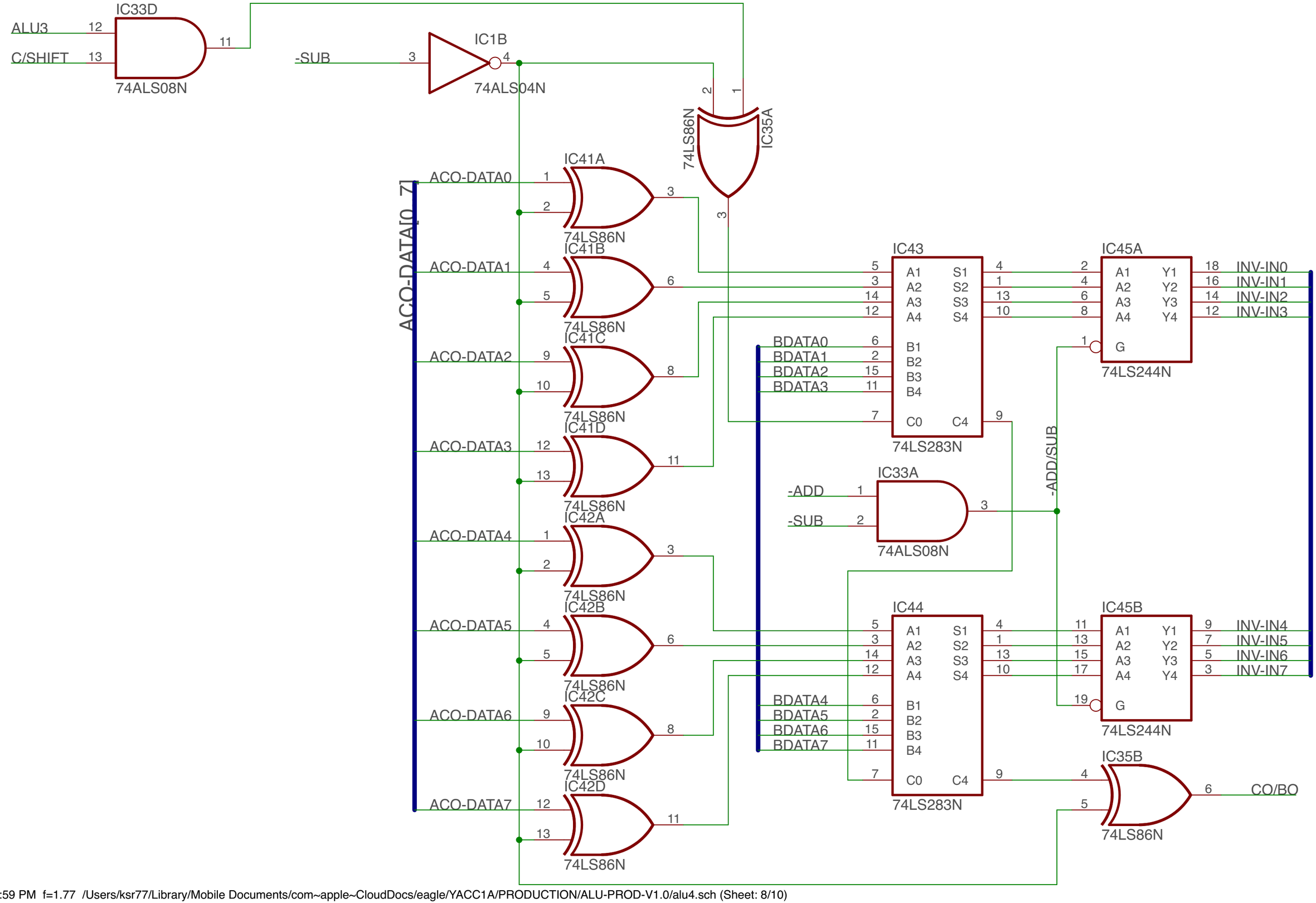

ACO-DATA[0..7]
BDATA[0..7]
IC21A
IC21B
IC21C
IC21D
IC22A
IC22B
IC22C
IC22D
74LS86N
IC24A
IC24B
74LS244N
-XOR
ACO-DATA0
BDATA0
ACO-DATA1
BDATA1
ACO-DATA2
BDATA2
ACO-DATA3
BDATA3
ACO-DATA4
BDATA4
ACO-DATA5
BDATA5
ACO-DATA6
BDATA6
ACO-DATA7
BDATA7
INV-IN0
INV-IN1
INV-IN2
INV-IN3
INV-IN4
INV-IN5
INV-IN6
INV-IN7

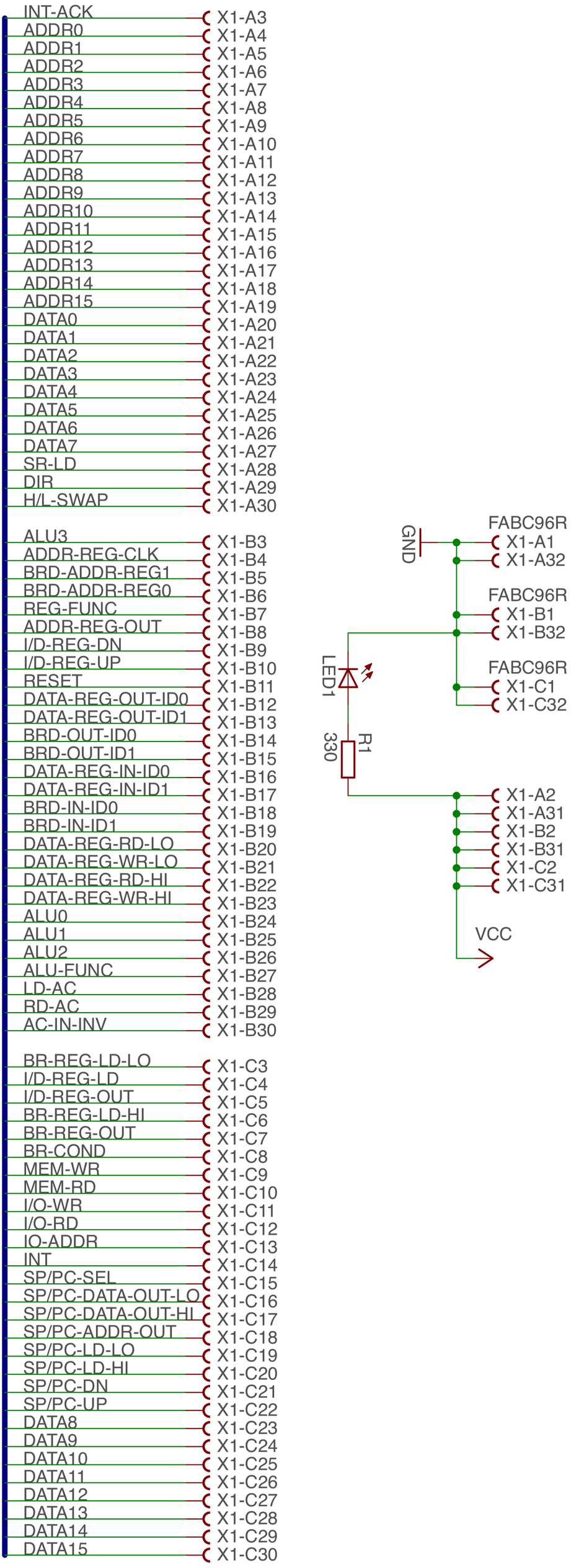

ACO-DATA[0..7]
BDATA[0..7]
IC25A
IC25B
IC25C
IC25D
IC26A
IC26B
IC26C
IC26D
74LS08N
IC27A
IC27B
74LS244N
-AND
ACO-DATA0
BDATA0
ACO-DATA1
BDATA1
ACO-DATA2
BDATA2
ACO-DATA3
BDATA3
ACO-DATA4
BDATA4
ACO-DATA5
BDATA5
ACO-DATA6
BDATA6
ACO-DATA7
BDATA7
INV-IN0
INV-IN1
INV-IN2
INV-IN3
INV-IN4
INV-IN5
INV-IN6
INV-IN7

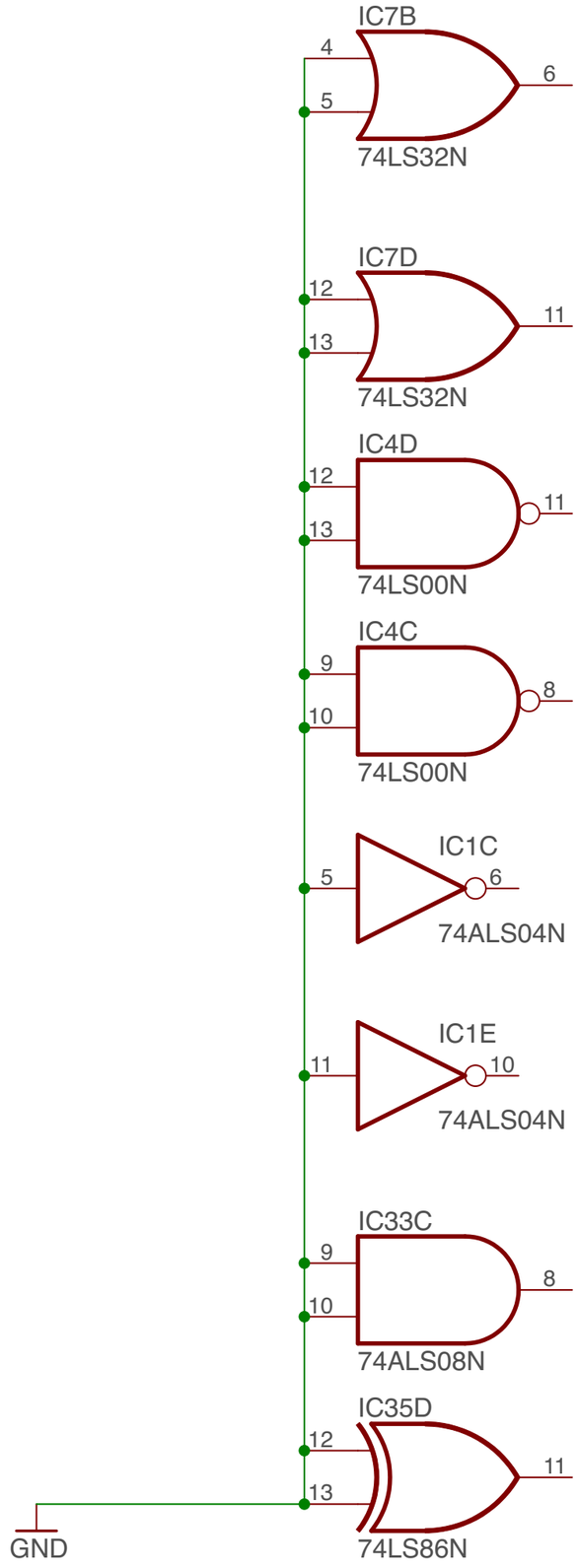

ACO-DATA[0..7]
BDATA[0..7]
IC28A
IC28B
IC28C
IC28D
IC29A
IC29B
IC29C
IC29D
74LS32N
IC30A
IC30B
74LS244N
-OR
ACO-DATA0
BDATA0
ACO-DATA1
BDATA1
ACO-DATA2
BDATA2
ACO-DATA3
BDATA3
ACO-DATA4
BDATA4
ACO-DATA5
BDATA5
ACO-DATA6
BDATA6
ACO-DATA7
BDATA7
INV-IN0
INV-IN1
INV-IN2
INV-IN3
INV-IN4
INV-IN5
INV-IN6
INV-IN7


IC31
BDATA0
BDATA1
BDATA2
BDATA3
ACO-DATA0
ACO-DATA1
ACO-DATA2
ACO-DATA3
ACO-DATA[0..7]
BDATA[0..7]
VCC
GND
A0
A1
A2
A3
B0
B1
B2
B3
A<B_I
A=B_I
A>B_I
A<B_O
A=B_O
A>B_O
74LS85N
IC32
BDATA4
BDATA5
BDATA6
BDATA7
ACO-DATA4
ACO-DATA5
ACO-DATA6
ACO-DATA7
AC-IN-INV
IC34
D0
D1
D2
D3
D4
D5
D6
D7
W
Y
A
B
C
G
ALU0
ALU1
ALU2
74LS251N
IC35C
74LS86N
BR-COND
C/SHIFT
V1
V1/+UB
V1/-UB
744078N
BDATA8
BDATA9
BDATA10
BDATA11
BDATA12
BDATA13
BDATA14
BDATA15
V2
V2/+UB
V2/-UB
IC33B
74ALS08N


IC36
1C0
1C1
1C2
1C3
2C0
2C1
2C2
2C3
1Y
2Y
A
B
1G
2G
74LS153N
ACO-DATA7
ACO-DATA0
ACO-DATA7
ALU2
ALU3
GND
C/SHIFT
GND
IC37
SR
A
B
C
D
SL
CLK
S0
S1
CLR
QA
QB
QC
QD
74LS194N
VCC
ACO-DATA0
ACO-DATA1
ACO-DATA2
ACO-DATA3
ACO-DATA[0..7]
IC38
ACO-DATA4
ACO-DATA5
ACO-DATA6
ACO-DATA7
SR-LD
ALU0
ALU1
74LS194N
VCC
IC39A
A1
A2
A3
A4
G
Y1
Y2
Y3
Y4
74LS244N
INV-IN0
INV-IN1
INV-IN2
INV-IN3
IC39B
INV-IN4
INV-IN5
INV-IN6
INV-IN7
74LS244N
-SHIFT
IC40
74LS153N
ALU0
ALU1
GND
VCC
IC8B
PRE
D
CLK
CLR
Q
Q
74LS74N
SHIFT-OUT


IC33D
ALU3
12
C/SHIFT
13
11
74ALS08N
IC1B
-SUB
3
4
74ALS04N
2
1
74LS86N
IC35A
3
ACO-DATA[0..7]
IC41A
ACO-DATA0
1
2
3
74LS86N
IC41B
ACO-DATA1
4
5
6
74LS86N
IC41C
ACO-DATA2
9
10
8
74LS86N
IC41D
ACO-DATA3
12
13
11
74LS86N
IC42A
ACO-DATA4
1
2
3
74LS86N
IC42B
ACO-DATA5
4
5
6
74LS86N
IC42C
ACO-DATA6
9
10
8
74LS86N
IC42D
ACO-DATA7
12
13
11
74LS86N
IC43
5
3
14
12
A1
A2
A3
A4
S1
S2
S3
S4
4
1
13
10
BDATA0
BDATA1
BDATA2
BDATA3
6
2
15
11
B1
B2
B3
B4
7
C0
C4
9
74LS283N
IC45A
2
4
6
8
A1
A2
A3
A4
Y1
Y2
Y3
Y4
18
16
14
12
INV-IN0
INV-IN1
INV-IN2
INV-IN3
1
G
74LS244N
-ADD/SUB
IC33A
-ADD
1
-SUB
2
3
74ALS08N
IC44
5
3
14
12
A1
A2
A3
A4
S1
S2
S3
S4
4
1
13
10
BDATA4
BDATA5
BDATA6
BDATA7
6
2
15
11
B1
B2
B3
B4
7
C0
C4
9
74LS283N
IC45B
11
13
15
17
A1
A2
A3
A4
Y1
Y2
Y3
Y4
9
7
5
3
INV-IN4
INV-IN5
INV-IN6
INV-IN7
19
G
74LS244N
IC35B
4
5
6
74LS86N
CO/BO


INT-ACK
X1-A3
ADDR0
X1-A4
ADDR1
X1-A5
ADDR2
X1-A6
ADDR3
X1-A7
ADDR4
X1-A8
ADDR5
X1-A9
ADDR6
X1-A10
ADDR7
X1-A11
ADDR8
X1-A12
ADDR9
X1-A13
ADDR10
X1-A14
ADDR11
X1-A15
ADDR12
X1-A16
ADDR13
X1-A17
ADDR14
X1-A18
ADDR15
X1-A19
DATA0
X1-A20
DATA1
X1-A21
DATA2
X1-A22
DATA3
X1-A23
DATA4
X1-A24
DATA5
X1-A25
DATA6
X1-A26
DATA7
X1-A27
SR-LD
X1-A28
DIR
X1-A29
H/L-SWAP
X1-A30
ALU3
X1-B3
ADDR-REG-CLK
X1-B4
BRD-ADDR-REG1
X1-B5
BRD-ADDR-REG0
X1-B6
REG-FUNC
X1-B7
ADDR-REG-OUT
X1-B8
I/D-REG-DN
X1-B9
I/D-REG-UP
X1-B10
RESET
X1-B11
DATA-REG-OUT-ID0
X1-B12
DATA-REG-OUT-ID1
X1-B13
BRD-OUT-ID0
X1-B14
BRD-OUT-ID1
X1-B15
DATA-REG-IN-ID0
X1-B16
DATA-REG-IN-ID1
X1-B17
BRD-IN-ID0
X1-B18
BRD-IN-ID1
X1-B19
DATA-REG-RD-LO
X1-B20
DATA-REG-WR-LO
X1-B21
DATA-REG-RD-HI
X1-B22
DATA-REG-WR-HI
X1-B23
ALU0
X1-B24
ALU1
X1-B25
ALU2
X1-B26
ALU-FUNC
X1-B27
LD-AC
X1-B28
RD-AC
X1-B29
AC-IN-INV
X1-B30
BR-REG-LD-LO
X1-C3
I/D-REG-LD
X1-C4
I/D-REG-OUT
X1-C5
BR-REG-LD-HI
X1-C6
BR-REG-OUT
X1-C7
BR-COND
X1-C8
MEM-WR
X1-C9
MEM-RD
X1-C10
I/O-WR
X1-C11
I/O-RD
X1-C12
IO-ADDR
X1-C13
INT
X1-C14
SP/PC-SEL
X1-C15
SP/PC-DATA-OUT-LO
X1-C16
SP/PC-DATA-OUT-HI
X1-C17
SP/PC-ADDR-OUT
X1-C18
SP/PC-LD-LO
X1-C19
SP/PC-LD-HI
X1-C20
SP/PC-DN
X1-C21
SP/PC-UP
X1-C22
DATA8
X1-C23
DATA9
X1-C24
DATA10
X1-C25
DATA11
X1-C26
DATA12
X1-C27
DATA13
X1-C28
DATA14
X1-C29
DATA15
X1-C30
GND
FABC96R
X1-A1
X1-A32
FABC96R
X1-B1
X1-B32
FABC96R
X1-C1
X1-C32
LED1
R1
330
X1-A2
X1-A31
X1-B2
X1-B31
X1-C2
X1-C31
VCC


IC7B
4
5
6
74LS32N
IC7D
12
13
11
74LS32N
IC4D
12
13
11
74LS00N
IC4C
9
10
8
74LS00N
IC1C
5
6
74ALS04N
IC1E
11
10
74ALS04N
IC33C
9
10
8
74ALS08N
IC35D
12
13
11
74LS86N
GND